REESHITA KATE

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SUMMARY

Computer Science student with a cybersecurity focus who builds and tests software, from web applications to machine learning models.

EDUCATION

Symbiosis Skills and Professional University | *Pune, India* August 2027 (Expected)

B.Tech in Computer Science and Information Technology (Cybersecurity) | CGPA: 8.9 / 10

TECHNICAL SKILLS

Languages: Python, Java, SQL (MySQL), C, HTML/CSS

Core CS: OOP, Data Structures & Algorithms, DBMS, Computer Networks

Backend & DevOps: REST APIs, Flask, SQLAlchemy, Docker, Git, GitHub Actions CI, pytest, Linux, AWS (EC2)

Security: VAPT, OWASP Top 10, Threat Modeling, Burp Suite Pro, OWASP ZAP

ML & AI: PyTorch, TensorFlow, Hugging Face Transformers, ONNX, pandas

EXPERIENCE

Trainee Software Engineer – Security | *Sciqus Infotech Pvt. Ltd., Pune* Jun 2026 – Jul 2026

- Carried out an independent VAPT engagement across the APIs of a multi-tenant production platform, delivering **60+ confirmed findings** rated by CVSS and mapped to the OWASP Top 10, validated in part via direct SQL queries.
- Uncovered a **systemic broken access control / IDOR flaw across nearly all API endpoints** through structured threat modeling, confirming account takeover, privilege escalation and session hijacking risks.
- Authored remediation-focused technical reports translating findings into prioritized, actionable fixes for engineering stakeholders, then **re-tested the patched issues** to confirm each fix held.
- Proved a **subdomain takeover** risk with a working proof of concept, using Burp Suite Pro, OWASP ZAP, subfinder and FFuF.

UI/UX and Web Development Intern | *Erfinden Technologies, Pune* Jun 2025 – Jul 2025

- Designed and launched a responsive, accessibility-first e-commerce site for a luxury retail client on Squarespace AI.

Remote Sensing Intern | *ARC Solutions, Pune* Jun 2024 – Jul 2024

- Built Python classifiers on Sentinel-1 SAR imagery in Google Earth Engine for flood detection and water-body mapping.

PROJECTS

VulnScan Pro – Web Vulnerability Scanner | *Python, Flask, SQLAlchemy, Docker, GitHub Actions, AWS* | GitHub

- Crawls a target website, runs **15 security checks** and grades it **A–F**, mapping every finding to OWASP Top 10, CWE and CVSS.
- Built the scanner modules, scoring engine and a Flask REST API on a SQL database, with authentication and rate limiting.
- Every push runs linting and **52 pytest tests** in CI; a backtest harness measures **true and false positives per module**.
- Deployed a public demo on **AWS EC2** with Docker, locked so it can only scan a bundled internal lab.

SAGE – Browser-Side DLP for Generative AI Prompts | *PyTorch, Hugging Face, ONNX*

- Chrome extension that blocks or redacts sensitive data in GenAI prompts using rules, fingerprinting and an ML classifier.
- Owned the ML classifier: **fine-tuned DistilBERT**, run in-browser via ONNX; **0.89 macro F1 vs 0.69** for the best baseline.

MedSentry – Anomaly Detection for Medical Device Networks | *Capstone, Python, HL7 v2, Docker*

- Hospital network monitor flagging HL7 device messages that are valid but clinically implausible, such as forged vitals.
- Owned the dataset and evaluation: a Python generator turning open ICU data into HL7 streams (**53 patients, 3,395 messages**) and a harness scoring the detector against **five baselines**.

Federated Learning-Based Intrusion Detection System for IoT Networks | *Personal project, TensorFlow*

- Trained with federated learning across three public IoT datasets (11M+ samples), so raw traffic never leaves the device.
- Multi-encoder DNN + AutoEncoder model reaching **over 90% accuracy on unseen environments vs 25–40%** for centralized models, with **under 1% false positives**.

CERTIFICATIONS

- Certified in Cybersecurity (CC) – ISC2
- AWS Security Fundamentals – AWS
- Security Operations Center in Practice – IBM
- Intermediate Ethical Hacker, Cybersecurity Fundamentals, Python Essentials – Cisco Networking Academy